REFERENCE GUIDE

The Binomial Distribution

Definitions, Formulas, and Worked Examples

This guide covers the binomial probability distribution as used in the BBE exam format: a fixed number of independent trials, each with the same probability of success. It presents the required conditions, the probability mass function, the mean and variance, the translation of exam wording into cumulative sums, the standard comparisons made between two distributions, common sources of error, and the normal approximation used for large samples. Each section includes a worked numerical example.

BBE School — Mathematics

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01 Definition

The binomial distribution describes the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes and the same probability of success. It is a discrete distribution: the random variable X, the number of successes, takes only integer values from 0 to n.

Success The outcome being counted, with fixed probability p on each trial — for example, a correctly answered question, a made shot, or a correctly identified defect.

Failure Any other outcome, with probability 1 − p on each trial. This is not evaluative; it denotes only the complement of the outcome being counted.

EXAMPLE 1.1

A multiple-choice exam has 10 questions, each with 4 answer choices and exactly one correct choice. A student guesses randomly on every question.

Trial	One question
“Success”	Answering correctly
n (number of trials)	10
p (probability of success)	0.25
Possible values of X	0 through 10

02 Required Conditions

A scenario must satisfy the following four conditions before the binomial formula can be applied.

Condition	Requirement	Example 1.1 check
Fixed n	The number of trials is determined in advance.	n = 10 questions, fixed before the exam is taken.
Binary outcome	Each trial results in exactly one of two outcomes.	Each question is either correct or incorrect.
Constant p	The probability of success is identical on every trial.	p = 0.25 on every question, since each has 4 choices.
Independence	The outcome of one trial does not affect another.	Guessing on question 3 does not affect the odds on question 4.

03 Probability Mass Function

The probability of obtaining exactly k successes in n trials is given by:

PROBABILITY MASS FUNCTION

$$P(X=k) = \frac{n!}{k!\,(n-k)!} \times p^{k}(1-p)^{n-k}$$

The term *C(n, k)*, read “n choose k,” equals the number of ways to select k successes among n trials, and is defined as:

$$C(n,k) = \frac{n!}{k!\,(n-k)!}$$

C(n, k) — combinatorial term The number of distinct orderings of k successes and (n − k) failures among n trials. It does not depend on which specific trials succeed, only on the count.

$p^k(1-p)^{n-k}$ **— probability term** The probability of any one specific ordering: k independent successes, each with probability p, and (n − k) independent failures, each with probability (1 − p).

EXAMPLE 3.1 — worked calculation

Using n = 10, p = 0.25 (Example 1.1), compute P(X = 3): the probability that exactly 3 of the 10 guesses are correct.

C(10, 3)	120
$p^3 = 0.25^3$	0.015625
$(1-p)^7 = 0.75^7$	0.133484
P(X = 3) = 120 × 0.015625 × 0.133484	0.2503

$$C(10,3) = \frac{10!}{3! \times 7!} = 120$$

Result: P(X = 3) ≈ 25.03%

04 Mean and Variance

The mean, variance, and standard deviation summarize the distribution without listing every individual probability.

MEAN

$$\mu = E(X) = n \times p$$

VARIANCE — STANDARD FORM

$$\sigma^2 = \mathrm{Var}(X) = n \times p \times (1-p)$$

VARIANCE — AS A SUM OVER OUTCOMES (GENERAL DEFINITION)

$$\sigma^2 = \sum (x-\mu)^2\, P(X=x)$$

This is the definition variance is built from for any discrete distribution. For a binomial variable it simplifies to np(1 − p); it is rarely computed this way in practice but clarifies what variance measures.

STANDARD DEVIATION

$$\sigma = \sqrt{n \times p \times (1-p)}$$

Key relationship: variance and distance from p = 0.5

For a fixed n, the product p(1 − p) is maximized at p = 0.5 and decreases as p moves toward 0 or 1. Consequently, variance is largest when p is closest to 0.5, regardless of whether p is otherwise “high” or “low.”

EXAMPLE 4.1 — worked calculation

Two students each answer 20 questions. Student A has p = 0.50 per question; Student B has p = 0.80 per question. Compare their means and variances.

Mean, Student A = 20 × 0.50	10
Mean, Student B = 20 × 0.80	16
Variance, Student A = 20 × 0.50 × 0.50	5.0
Variance, Student B = 20 × 0.80 × 0.20	3.2

Conclusion: Student A’s scores vary more, despite the lower mean.

05 Cumulative Probabilities and Exam Wording

Statements in this exam format frequently describe a range of outcomes in words rather than in mathematical notation. The following table gives the standard translation. The translation depends only on the wording, not on the values of n or p.

Wording	Notation
“at least k”	Σ P(X = x), x = k to n
“at most k”	Σ P(X = x), x = 0 to k
“more than k”	Σ P(X = x), x = k+1 to n
“fewer than k”	Σ P(X = x), x = 0 to k−1
“between a and b, inclusive”	Σ P(X = x), x = a to b
“exactly k”	P(X = k), a single term

EXAMPLE 5.1 — worked calculation

Using n = 10, p = 0.90 (a student who studies well), compute the probability of passing an exam requiring at least 7 of 10 correct.

$P(X = 7)$	0.0574
$P(X = 8)$	0.1937
$P(X = 9)$	0.3874
$P(X = 10)$	0.3487
$P(X \geq 7)$ = sum of the above	0.9872

Result: $P(X \geq 7) \approx 98.72\%$

Percentage thresholds

When a threshold is expressed as a percentage of n, it frequently does not correspond to a whole number. For example, 90% of 18 units is 16.2 units. Since the count of successes must be an integer, “at least 90%” requires rounding 16.2 up to 17 (the smallest integer count that still meets the requirement). Conversely, “at most” thresholds round down to the nearest integer satisfying the cap. Rounding in the wrong direction changes the resulting summation range.

06 Comparing Two Distributions

Many exam questions present two entities with a shared binomial structure but different values of p, and ask for a comparison. The comparison generally takes one of the following four forms.

Ratio “B is X times as likely as A.” Compute both probabilities in full, then divide. The result should not be rounded to the nearest whole number before comparing to the stated threshold.

Difference “B exceeds A by more than X.” Subtract directly. For two means with the same n, the difference simplifies to $n \times (p_B - p_A)$.

Percentage increase “B is X% higher than A.” Compute the ratio of the two means, subtract 1, and multiply by 100. When n is shared, this ratio equals the ratio of the two p-values directly.

Variance or standard deviation ratio Compute $n \times p \times (1 - p)$ separately for each entity before dividing. Take the square root only if the comparison concerns standard deviation rather than variance.

EXAMPLE 6.1 — worked calculation

Using Example 4.1 (n = 20 for both students), compute the percentage by which Student B’s mean exceeds Student A’s mean.

Mean, Student A	10
Mean, Student B	16
Percentage increase, (ratio − 1) × 100	60%

$$\frac{16}{10} = 1.6$$

07 Common Errors

The following errors account for the majority of incorrect answers on this exam format.

1. Estimating ratio thresholds instead of computing them. A claim such as "more than 20 times as likely" is frequently placed close to the true ratio (for example, 19.7 or 20.3). Estimating rather than computing the exact ratio produces the wrong answer in these cases.

2. Assuming a higher mean implies a higher variance. The entity with the higher success probability (further from p = 0.5) frequently has the lower variance. Variance depends on distance from 0.5, not on which value of p corresponds to better performance.

3. Overlooking the symmetry between p and 1 − p. A binomial distribution with parameter p has the same variance, and the same magnitude of skewness (opposite sign), as one with parameter 1 − p, for the same n. Additionally, P(X = 0) at probability p equals P(X = n) at probability 1 − p.

4. Treating the sum of two different binomial variables as binomial. If two independent processes have different success probabilities, their combined count does not follow a binomial distribution with a pooled probability, even though the means still add. The true combined variance will generally not equal the variance of a pooled binomial with the same mean.

5. Checking only one side of the normal-approximation rule of thumb. The rule requires both n × p ≥ 5 and n × (1 − p) ≥ 5. A large value on one side does not compensate for a small value on the other.

08 Normal Approximation

For large n, computing the exact binomial probability term by term becomes impractical. The normal approximation is used, subject to a validity condition.

Normal approximation

Condition:

$$n \times p \geq 5 \quad \text{and} \quad n \times (1 - p) \geq 5$$

Continuity correction: for "at most k," use (k + 0.5); for "at least k," use (k − 0.5), before standardizing.

09 Summary Reference

Task	Method
Verify the distribution is binomial	Confirm fixed n, binary outcome, constant p, independence.
"Exactly k"	C(n,k) × p^k × (1−p)^(n−k)
"At least / at most / between"	Convert to a summation range per Section 05.
Percentage threshold given	Convert to a whole-number count; round up for "at least," round down for "at most."
"How many times as likely"	Compute both probabilities in full, then divide.
Which distribution has greater spread	The p-value closer to 0.5 yields the greater variance, for equal n.

Task	Method
Large n, approximation needed	Normal approximation if $n \times p \geq 5$ and $n \times (1-p) \geq 5$.

Binomial Distribution — Reference Guide, BBE School Mathematics.